SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE CODE: MLA03
COURSE NAME: Reinforcement learning for Environment and Agent
NAME: M.Madhulika
REGISTER NUMBER: 192472188

COURSE OUTCOME COVERED

CO3: *Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3, BL6).*

Assessment Tool Weightage: 25%

Dr G. A. SENTHIL

QUESTIONS: 10 **Total Marks: 50**

Annexure A

Data Interpretation

1. **A Reinforcement Learning agent is trained for multiple episodes and the reward values increase gradually from 10 to 45. Interpret the trend in rewards and defend what indicates about the learning performance and policy improvement of the agent.**

Solution: -

Learning Performance and Policy Improvement in Reinforcement Learning.

Data Observation

- The reward increases from **10 to 45** over multiple episodes.
- The increase is gradual and consistent.

Analysis

- The increasing reward shows that the agent is learning from its interactions with the environment.
- The agent is selecting better actions over time.

- The policy is improving with each training episode.

Interpretation

- Higher rewards indicate better decision-making.
- The agent is moving toward an optimal policy.
- Continuous learning helps improve overall performance.

Conclusion

The gradual increase in reward from **10 to 45** indicates that the RL agent is learning successfully. The improved rewards show better policy updates, effective learning, and progress toward an optimal solution.

2. An epsilon-greedy agent is tested with different epsilon values such as 0.1, 0.3 and 0.5. The cumulative reward obtained is highest when epsilon is 0.1 and lowest when epsilon is 0.5. Justify and apply the concept of exploration vs exploitation to interpret why the reward changes with epsilon and identify which epsilon value provides better performance.

Solution: -

Exploration vs. Exploitation using the ε-greedy strategy.

Data Observation

- Epsilon values tested: 0.1, 0.3, and 0.5.
- The highest cumulative reward is obtained with $\varepsilon = 0.1$.
- The lowest cumulative reward is obtained with $\varepsilon = 0.5$.

Analysis

- A lower epsilon (0.1) focuses more on exploitation, selecting the best-known action most of the time.
- A higher epsilon (0.5) increases exploration, causing more random actions and reducing rewards.

Interpretation

- $\varepsilon = 0.1$ provides the best balance between exploration and exploitation.
- The agent makes better decisions and achieves higher cumulative rewards.
- Excessive exploration reduces learning efficiency.

Conclusion

The results show that **$\varepsilon = 0.1$** gives the best performance because it balances exploration and exploitation effectively. Higher epsilon values lead to more random actions and lower cumulative rewards, making learning less efficient.

3. **Two Reinforcement Learning algorithms, Q-Learning and SARSA, are compared based on average rewards obtained over 100, 200 and 300 episodes. Q-Learning consistently shows slightly higher reward values than SARSA. Identify the performance difference between these algorithms and determine which algorithm converges faster based on the observed results.**

Solution: -

Comparison of Q-Learning and SARSA based on reward and convergence.

Data Observation

- Rewards are compared over 100, 200, and 300 episodes.
- Q-Learning consistently achieves slightly higher rewards than SARSA.

Analysis

- Higher rewards indicate that Q-Learning learns a better policy.
- SARSA learns more cautiously because it follows the current policy.
- Q-Learning reaches higher rewards in fewer episodes.

Interpretation

- Q-Learning converges faster than SARSA.
- It finds the optimal policy more quickly and achieves better performance.
- SARSA provides stable learning but converges more slowly.

Conclusion

The results show that Q-Learning outperforms SARSA by achieving higher average rewards and faster convergence. Therefore, Q-Learning is more efficient for this scenario, while SARSA is more conservative and stable during learning.

4. **An RL agent is trained using different discount factor values such as 0.2, 0.5 and 0.9. The average reward increases as the discount factor increases. Discuss how the discount factor affects long-term reward optimization and justify which discount factor is most suitable for achieving better learning performance.**

Solution: -

Discount Factor (γ) and its effect on long-term reward optimization.

Data Observation

- Discount factors tested: 0.2, 0.5, and 0.9.
- The average reward increases as the discount factor increases.
- $\gamma = 0.9$ gives the highest reward.

Analysis

- A low discount factor (**0.2**) focuses mainly on immediate rewards.
- A high discount factor (**0.9**) considers future rewards, leading to better long-term decisions.
- Higher γ helps the agent learn a more effective policy.

Interpretation

- **γ = 0.9** provides the best learning performance.
- The agent balances immediate and future rewards effectively.
- This results in higher cumulative rewards and improved policy optimization.

Conclusion

The results show that a discount factor of 0.9 is the most suitable because it emphasizes long-term rewards, improves learning performance, and helps the agent converge toward an optimal policy.

5. The success rate of an RL agent increases from 45 percent at 50 episodes to 92 percent at 200 episodes. Discuss the improvement in success rate and interpret whether the agent has learned an optimal policy. Provide reasoning based on the trend of the data.

Solution: -

Policy Learning and Success Rate in Reinforcement Learning.

Data Observation

- Success rate increases from 45% at 50 episodes to 92% at 200 episodes.
- The success rate improves consistently with more training.

Analysis

- The increasing success rate indicates that the agent is learning from experience.
- More training episodes help the agent improve its decision-making.
- The learned policy becomes more effective over time.

Interpretation

- A 92% success rate suggests the agent has learned a near-optimal policy.
- The agent makes correct decisions more frequently.
- The continuous improvement reflects effective learning and policy optimization.

Conclusion

The increase in success rate from 45% to 92% shows that the RL agent has learned successfully through repeated interactions. The trend indicates significant policy improvement and suggests that the agent is approaching an optimal policy with high learning performance.

Code of Conduct:

I M.Madhulika (192472188) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
M.Madhulika

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand